

EXCEL:DATA CLEANING AND

EXPLORATION BRIEF EXPLANATION:-

1.)HANDLING MISSING VALUES:-

Selected Product ID as Main Preference Column to remove Duplicate rows within the dataset based on the entirety of each row & clicked on Data>Data Tools>Remove Duplicates, When pop-up stating "remove duplicates" opened, so selected "Expand the row", also checked on "Row containing Headers" & Removed 3 Duplicate Product ID's - "17-JUN-IN" , "16-APR-ES" and "21-AUG-CA" , So found 31 unique Product IDs on the entirety of each row in the dataset.

I Handled Products with Missing Price Information in "Price" Column with empty cells - I7,I21 & I27 on Rows - 7,21 & 27 using Statistical Imputation on Excel "IF ISBLANK" formula :
IF(ISBLANK(D7),AVERAGE(\$D\$2:\$D\$32),D7) & also using "AVERAGE IF" formulae : IF(ISBLANK(D7),AVERAGE IF(K\$2\$:K32,K7,\$D\$2:\$D\$32),D7) on Column with empty cells - L7,L21 & L27 on Rows - 7,21 & 27 .

I Handled Products with Missing Category Information in "Category" Column on empty cells - J6,J13,J14 & J24 on Rows - 6,13,14 & 24 using Constants or Default Values Such as "Unknown" also called categorical imputation on Excel using IF ISBLANK formula : IF(IS BLANK(F6),"Unknown",F6) .

2.)CORRECTING INCONSISTENT DATA:-

Inconsistencies found in "Product Name" Column Where the 1st letter in Some of the Product names such as "Laptop" in Rows - 2

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& 19 , "Smartphones" in Row - 5 & "Headphones" in Rows - 16 & 29 was in "LOWER CASE" . So Standardized text format to "UPPER CASE" Letters using "Find and Replace" function in Excel.

Also Found Typos or Misspelling for the name "Electroni" in "Category" Column specifically on Rows - 7,15,16,18 & 29. So Using "Find and Replace" function by Clicking on keyboard shortcut "CTRL+F" in Excel & Typed Name "Electroni" on "Find What" Search Box & Typed Name "Electronics" on "Replace With" search box & clicked on "Replace All" to Correct all incorrect Spelling of name "Electroni" to Correct Spelling of the name "Electronics" .

3.)REMOVING DUPLICATES:-

Selected Product ID as Main Preference Column to remove Duplicate rows within the dataset based on the entirety of each row & clicked on Data>Data Tools>Remove Duplicates, When pop-up stating"remove duplicates" opened, so selected "Expand the row",also checked on"Row containing Headers" & Removed 3 Duplicate Product ID's - "17-JUN-IN" , "16-APR-ES" and "21-AUG-CA" , So found 31 unique Product IDs on the entirety of each row in the dataset.

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4.)SPLITTING AND MERGING DATA:-

I Merged the "Brand Name" and "Product Name" Columns into a single Column named "Product Brand" From Cells I2 to I16 using "CONCATENATE" Function in Excel : CONCATENATE(D2 " " E2) and from cells I17 to I35 using "Ampersand (&)" function in excel : D18 & " " & E18 .

I Splitted the "Product ID" Column into Two separate columns as "Manufacturing Date" using the In-built Excel formulae : LEFT(A2,LEN(A2)-3) & "Country Code" using the In-built Excel formulae : RIGHT(A2,LEN(A2)-7) .

5.)NUMBER FORMATTING:-

Formatted "Column b" - "Manufacturing date into "DD-MM-YYYY" format by using the keyboard function "CTRL+1" , then changed Text Format "28-JAN" to Number Format using the year : 2026 by using excel formulae "TEXT(B2, "DD-MM-YYYY)" on "Column K" & got DATE : 28-01-2026 & also by using the year : 2025 with excel formulae "DATEVALUE(B2 & "-2025)" on "Column L" & got DATE : 28-01-2025 . So Applied Date Formats "DD-MM-YYYY" on all the rows of Columns B,L & K & So Formatted all the "Manufacturing Dates" Accordingly .

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So Formatted the data type of the "Price Column J" to "Currency Format" by Selecting entire "Price Column J" & by selecting "Currency" function on the ribbon called "Number" in excel. So Applied "Currency Formats" from "1000" to "1000.00" on all the rows of Price Column J & so Formatted Accordingly.

6.)CONDITIONAL FORMATTING:-

Used "Conditional Formatting" function on ribbon called "Styles" in excel, So Applied "Purple Colour" Data Bar on "Price Column D" , "Orangeish Yellow Colour" Data Bars on "Price Columns L and M" & used "Light and Dark orange" colour scales on "Price Columns I and J" .

Used "Conditional Formatting" function on ribbon called "Styles" in excel , So Applied custom rule for conditional formatting in the "Category" - "F" Column & used the function "Highlight Cell Rules" > "Text that Contains" to highlight cells where the category is "Electronics" by using Colour scale "Light Red fill with Dark Red Text" .